

CURRICULUM VITAE

NAME: HUONG THI THANH HA

POSITION TITLE: Lecturer (similar to Assistant Professor role in the US system)

EDUCATION/TRAINING

Institution and Location	Degree	Completion Date	Field of Study
EBRAINS & IBRO		7/2021	Neuroscience (First Virtual Master Class on Brain Atlas and Simulation Services)
France – Vietnam training for neurologists	Certificate	12/2019	Neuroscience
University of Medicine and Pharmacy, Ho Chi Minh City, Vietnam	Certificate	10/2019	Diagnostic test standard and procedure at hospitals.
Stanford University School of Medicine, Stanford, CA, US	Ph.D.	8/2018	Neuroscience
The Marine Biological Laboratory, MA, US	Certificate	8/2014	Neurobiology
University of Science, Vietnam National University, Ho Chi Minh City (HCMC VNU), Vietnam	B.S.	09/2011	Medical biotechnology

A. Personal Statement and Research Interest

Due to a family history of multiple members with mental disorders, I became intrigued by how the brain works, what molecules drive its functions, and how they changed when it gets dysregulated. During college, I studied molecular markers for devising diagnostic tests of cervical cancer. Following graduation, I further pursued this interest by working at the Oxford University Clinical Research Unit and investigating peripheral blood markers of HIV-associated neurological disorders (HAND). At Stanford University, my PhD research was focused on understanding the molecular system related to autism and cellular and network deficits underlying epilepsy. I also created an efficient gene delivery technology and a universal brain tissue culture platform for research and drug discovery applications.

Upon coming back to Vietnam and joining the International University as a lecturer, I established the Brain Health Lab and pursue several research projects to address mental health issues and cognitive-aging disorders in Vietnam. In terms of brain health, my primary focus is on enhancing diagnoses and interventions for dementia at the early stage. My research encompasses a range of aspects, including molecular diagnosis and AI applications. For molecular detection, I am leading early diagnosing AD, focusing on detecting p-Tau217 and cell-free RNA detection. For studies related to AI, I am developing an easy-to-use AI system to detect early signs of Alzheimer's Disease (AD) in clinical settings or the biological characteristics of the heterogeneity in the Mild Cognitive Impairment (MCI) population. For the intervention aspect, I am building a cognitive game for MCI patients to hinder their conversion to dementia and evaluate the efficacy of EEG signals in predicting post-intervention effectiveness. Besides dementia research, my studies also apply AI to build a Brain-Computer Interface system for stroke rehabilitation and investigate AI potential in detecting brain tumors and migraines. Along with those studies, I am leading projects to build the first Vietnamese databases of neuroscience-related data, such as brain MRI and EEG signals to enhance AI models and improve diagnosis. In the realm of mental health, my research delves into the relationship between stress and emotions based on changes in EEG signals, as well as the efficacy of techniques such as Alpha music and Mindfulness-Based Stress Reduction in reducing stress. Moreover, I am also looking into developing a low-cost, non-invasive point-of-care device to quantify cortisol for stress diagnosis. My goal is to create innovative, highly accurate, and low-cost products that can have a tangible impact on the brain health and mental well-being of those in my community and beyond.

B. Publications, Laboratory products, Presentations and Posters

Publication

2025

Feasibility of home blood pressure telemonitoring in older adults with Mild Cognitive Impairment and uncontrolled hypertension in Vietnam (Authors: Vy Le Tran, Tran Tran To Nguyen, The Ha Ngoc Than, Dinh Chau Bao Hoang, Tuan Chau Nguyen, Viet Ngoc Tran, Toi Van Vo, Nam Phuong Nguyen, **Huong Thi Thanh Ha**) (Accepted, Biomedical Research and Therapy)

Assessing the feasibility of cognitive intervention via BrainTrain cognitive gaming application for Mild Cognitive Impairment patients (Authors: Thu Ngoc Minh Phan, Binh Nguyen Le Nguyen, Bao Quoc Nguyen, Nhung Hoang Do, Nhu Thi Huynh Tran, Nhan Thanh Nguyen, Thu Hoang Anh Pham, Nhan Quoc Trung Nguyen, Tuong Huu Nguyen, Kien Trung Duong, Nghia Tien Trong Hoang, Thu Thi Hoai Tran, Dang Minh Ly, and Dieu Xuan Nguyen Tan Le-Duy, and **Huong Ha Thi Thanh**) (Accepted, Int. J. Bioinformatics Research and Applications)

Ma Thi, C., Kien Nguyen, M., Vu Thanh, L., Nguyen Dinh, H., Huynh Thi, N. Y., **Ha, H. T.**, ... & Le Hoang, T. L. UET175: EEG dataset of motor imagery tasks in Vietnamese stroke patients. *Frontiers in Neuroscience*, 19, 1580931.

Ngo, L., Le, A., Ho, K., Le, T., Nguyen, N., Nguyen, S., ... & **Ha, H.** (2025). Electroencephalogram (EEG) based automated detection of mental disorders using artificial intelligence processing pipelines. *Neuroscience Research Notes*, 8(2), 404-1.

Tran, V. Q. T., Tran, N. T. L., Nguyen, L. Q., An, H. M. A., Nguyen, T. H., Trang, H. L., ... & **Ha, T. T. H.** (2025). Effect of Alpha Music on the Stress Level of Vietnamese Students. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 363-375). Springer, Cham.

Nguyen, T. M. H., Thi, T. M. N., **Ha, H. T. T.**, Truong, N. T. N., Nguyen, T. H., Cong, C. P., & Ngoc, T. M. P. (2025). Multi-stage Alzheimer's Diagnosis Using MRI Image: A Combination of Computer Vision Technique for Machine Learning Model. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 376-393). Springer, Cham.

Nguyen, X. K., Ngo, T. A., Nguyen, T., Pham, P., **Ha, H.**, & Ngo, L. (2025). A Comparative Study of Transfer Learning Techniques with BERT Family Models in Early Diagnosis of Alzheimer's Disease. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 373-382). Springer, Cham.

Luu, T., Dang-Hieu, A., Ly, T., Pham, H. T., **Ha, H.**, & Ngo, L. (2025). Developing a Photoplethysmography-Based Automated Sleep-Stage Scoring Framework Using Artificial Intelligence. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 394-408). Springer, Cham.

Thanh, N. N. T., Thanh, **H. H. T.**, **Thanh**, H. P., Thi, L. N., Trong, N. H. T., & Hoai, T. T. T. (2025). Recent Approaches on Acoustic and Language-Based Analysis in Alzheimer's Disease: A Literature Review. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 459-491). Springer, Cham.

Huynh, V., Nguyen, Q., Do, H., Le, A., Vo, T., & **Ha, H.** (2025). An Assessment of Alpha Music Effects on Brain-Computer Interface Performance: A Preliminary Study. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 555-575). Springer, Cham.

Nguyen, M., Duong, N. V. B., Duc, V. T., Phan, C., Phan, T., & **Ha, H.** (2025). Unveiling Heterogeneity in Early Alzheimer's Disease: A Data-Driven Exploration of Mild Cognitive Impairment and Normal Cognition. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 408-427). Springer, Cham.

2024

Chu, L. H., Chau, C. Q., Kamel, N., **Thanh, H. H. T.**, & Yahya, N. (2024). Functional excitation-inhibition ratio for social anxiety analysis and severity assessment. *Frontiers in Psychiatry*, 15, 1461290.

Nguyen, H. D., Huynh, T., Nguyen, M., Nguyen, H., Bui, L. C., Ta, T. M., ... & **Ha, H.** (2024). Comparing the Hippocampal Volume Between Patients with Alzheimer's Disease and Healthy Controls: An MRI Analysis using the Harmonized Hippocampal Protocol. *Brain Science Advances*, 10(4), BSA-2024.

Van Tran, L., Le, A. P. H., Le, T. T. M., Nguyen, M. T. T., Nguyen, Q. L., & **Ha, H. T. T.** (2024). Comparing the efficacy of online vs. offline methods in modulating emotional responses through mental stress tasks. *Vietnam Journal of Science, Technology and Engineering*, 66(2), 107-113.

Nguyen, M., Pham, B., Vo, T., & **Ha, H.** (2024). Revealing the heterogeneity of plasma protein and cognitive decline trajectory among Mild Cognitive Impairment patients by clustering of brain atrophy features. *Brain Multiphysics*, 100093.

Trong, K. N., Luong, N. N., Tan, H., Trung, D. T., Thanh, **H. H. T.**, & Thanh, B. N. (2024). Real-time Single-Channel EOG removal based on Empirical Mode Decomposition. *EAI Endorsed Transactions on Industrial Networks and Intelligent Systems*, 11(2).

2023

Nguyen, K., Nguyen, M., Dang, K., Pham, B., Huynh, V., Vo, T., ... & **Ha, H.** (2023). Early Alzheimer? s disease diagnosis using an XG-Boost model applied to MRI images. *Biomedical Research and Therapy*, 10(9), 5896-5911.

Duong, H., Nguyen, T., Pham, D., Duong, K., Mieu, L., Nguyen, H., ... **Ha, H. T. T.** & Ngo, L. (2023, December). HosRev: A Comprehensive Aspect-Category Sentiment Analysis Dataset on Hospital Reviews in Vietnam. In *2023 1st International Conference on Health Science and Technology (ICHST)* (pp. 1-6). IEEE.

Do, T. T., Van Tran, L., Le, T. A., Le, T. M. T., Duong, L. A. H., Nguyen, T. H., ... & **Ha, H. T. T.** (2023, December). Stress Prediction Using Machine-Learning Techniques on Physiological Signals. In *2023 1st International Conference on Health Science and Technology (ICHST)* (pp. 1-7). IEEE.

Luu, T., Luu, T., Bui, T. T., Trieu, C., Truong, D., Hoang, N., ... & **Ha, H.** (2023, December). VHCOR: Vietnamese Healthcare Corpus-A Comprehensive Dataset for Vietnamese Medical Department Recognition. In *2023 1st International Conference on Health Science and Technology (ICHST)* (pp. 1-6). IEEE.

Cao, T. H. M., Le, A. P. H., Tran, T. T., Huynh, V. K., Pham, B. H., Le, T. M., ... & **Ha, H. T. T.** (2023). Plasma cell-free RNA profiling of Vietnamese Alzheimer's patients reveals a linkage with chronic inflammation and apoptosis: a pilot study. *Frontiers in Molecular Neuroscience*, 16.

Phan, T. N. M., Nguyen, H. T., Huynh, T. N. M., Nguyen, T. H., Tran, T. N. Y., & **Ha, H. T. T.** (2022, December). Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly. In *International Conference on the Development of Biomedical Engineering in Vietnam* (pp. 719-737). Cham: Springer Nature Switzerland.

Le, A. P. H., Nguyen, Q. L., Pham, B. H., Cao, T. H. M., Van Vo, T., Huynh, K., & **Ha, H. T. T.** (2023). SALAD: Syringe-based Arduino-operated Low-cost Antibody Dispenser. *HardwareX*, e00455.

Nguyen, T. T. Q., Hoang, C. B. D., Hoang Le, M. D., Anh Vo, N. T., Quang, H., Nguyen, C. M., ... & **Thanh Huong Ha, T.** (2023). Assessing cognitive decline in Vietnamese older adults using the Montreal Cognitive Assessment-Basic (MoCA-B) and Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE) during the COVID-19 pandemic: a feasibility study. *The Clinical Neuropsychologist*, 1-19.

Nguyen, M. T. D., Xuan, N. Y. P., Pham, B. M., Do, H. T. M., Phan, T. N. M., Nguyen, Q. T. T., ... & **Ha, H. T. T.** (2023). Optimize temporal configuration for motor imagery-based multiclass performance and its relationship with subject-specific frequency. *Informatics in Medicine Unlocked*, 36, 101141.

Do, H., Hoang, H., Nguyen, N., An, A., Chau, H., Khuu, Q., ... & **Ha, H.** (2023). Intermediate effects of mindfulness practice on the brain activity of college students: An EEG study. *IBRO Neuroscience Reports*, 14, 308-319.

2022

An, A., Hoang, H., Trang, L., Vo, Q., Tran, L., Le, T., ... & **Ha, H.** (2022). Investigating the effect of Mindfulness-Based Stress Reduction on stress level and brain activity of college students. *IBRO neuroscience reports*, 12, 399-410.

Dang, K., Vo, T., Ngo, L., & **Ha, H.** (2022). A deep learning framework integrating MRI image preprocessing methods for brain tumor segmentation and classification. *IBRO neuroscience reports*, 13, 523-532.

Nguyen, L. Q., An, H. M. A., Tran, V. Q. T., Le, P. H. A., Le, M. T., Nguyen, T. K., & **Ha, T. T. H.** (2022). Investigation of the effect of alpha wave-containing music on the cognitive performance and brain activity of university students. *Vietnam Journal of Science, Technology and Engineering*, 64(3), 69-75.

Nguyen, D., Nguyen, H., Ong, H., Le, H., **Ha, H.**, Duc, N. T., & Ngo, H. T. (2022). Ensemble learning using traditional machine learning and deep neural network for diagnosis of Alzheimer's disease. *IBRO Neuroscience Reports*, 13, 255-263.

Ngan Thanh, L., Anh Hoang Lan, D., Dung Xuan, N., Khiết Thi Thu, D., Chau Nu Ngoc, P., Thuong Hoai, N., & **Huong, H. T. T.** (2022). Classification of Four-Class Motor-Imagery Data for Brain-Computer Interfaces. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 763-778). Springer International Publishing.

Do, T. T., Nguyen, T. H., Le, T. A., Nguyen, S. A. T., Nguyen, Q. T. N., Tran, T. Q. V., ... & **Ha, H. T. T.** (2022). Automated EOG removal from EEG signal using independent component analysis and machine learning algorithms. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 1001-1016). Springer International Publishing.

Van Toi, V., Nguyen, T. H., Long, V. B., & **Huong, H. T. T.** (2022). 8th International Conference on the Development of Biomedical Engineering in Vietnam.

Ong, H., Le, H., Nguyen, H., Nguyen, D., **Ha, H.**, Ngo, H. T., & Duc, N. T. (2022). A Machine Learning Framework Based on Extreme Gradient Boosting for Intelligent Alzheimer's Disease Diagnosis Using Structure MRI. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 815-827). Springer International Publishing.

Anh Hoang Minh, A., Nghia Trung, N., Thuong Hoai, N., Khiết Thi Thu, D., & **Ha, H. T. T.** (2022). The Effect of Mindfulness-Based Stress Reduction Among Various Subject Groups: A Literature Review. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 869-899). Springer International Publishing.

Le, A. P. H., Tran, T. T., Cao, T. H. M., Le, T. M., Le, P. T., & **Huong, H. T. T.** (2022). Evaluate and Optimize Cell-Free RNA Extraction Methods to Apply for Alzheimer's Disease Biomarkers Detection. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 591-609). Springer International Publishing.

Le, A. P. H., & **Huong, H. T. T.** (2022). The Role of Cell-Free RNA in Clinical Diagnosis and Evaluation of Cell-Free RNA Extraction Methods. In *8th International Conference on the Development of Biomedical Engineering in Vietnam: Proceedings of BME 8, 2020, Vietnam: Healthcare Technology for Smart City in Low-and Middle-Income Countries* (pp. 637-656). Springer International Publishing.

Nguyen, T. D., & **Ha, T. T. H.** (2022). Bipolar disorder traits: An electroencephalogram systematic review. *Vietnam Journal of Science, Technology and Engineering*, 64(4), 84-90.

2021

Thinh, T. V. Q., Quang, N. L., Anh, A. H. M., Long, T. H., Hieu, N. T., Van Luan, T., ... & **Huong, H. T. T.** (2021, May). Effect of Alpha Wave-Containing Music on the Psychological Stress Level of Hospital Staffs and their Cognitive Performance. In *2021 IEEE 3rd Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (ECBIOS)* (pp. 25-28). IEEE.

2020

Fujita, H., ... & **Ha, H.** (2020, September). SEED: A Framework for Stress Estimation Using Emotiv Devices. In *Knowledge Innovation Through Intelligent Software Methodologies, Tools and Techniques: Proceedings of the 19th International Conference on New Trends in Intelligent Software Methodologies, Tools and Techniques (SoMeT_20)* (Vol. 327, p. 435). IOS Press.

Pham, V. M., **Ha, H. T.**, & Thakor, N. (2020). Microfluidic Culture Platforms in Neuroscience Research. *Handbook of Neuroengineering*, 1-39.

Tran, N. M. P., Huynh, T. L. D., Phan, B. N., Dang, N. N. T., Phan, T. B., **Ha, H. T. T.**, ... & Nguyen, T. H. (2020). Conjugated linoleic acid grafting improved hemocompatibility of the polycaprolactone electrospun membrane. *International Journal of Polymer Science*, 2020, 1-8.

Tran, N., Le, A., Ho, M., Dang, N., **Thi Thanh, H. H.**, Truong, L., ... & Hiep, N. T. (2020). Polyurethane/polycaprolactone membrane grafted with conjugated linoleic acid for artificial vascular graft application. *Science and Technology of Advanced Materials*, 21(1), 56-66.

2018

Ha, H. T., Leal-Ortiz, S., Lalwani, K., Kiyonaka, S., Hamachi, I., Mysore, S. P., ... & Kim, S. A. (2018). Shank and zinc mediate an AMPA receptor subunit switch in developing neurons. *Frontiers in molecular neuroscience*, 11, 405.

2016

Ha, H., & Huguenard, J. (2016). Criminal Minds: Cav3. 2 Channels Are the Culprits, but NMDAR Are the Co-Conspirators. *Epilepsy Currents*, 16(1), 36-38.

Laboratory products

BrainTrain

<https://www.braintrainhcmiu.com/>

https://www.facebook.com/profile.php?id=100093556011720&_rdr

BrainConnects 2023

<https://brainconnects2023.wixsite.com/brain-connects>

https://www.facebook.com/BrainConnects/?_rdr

BrainAnalytics

<https://brainanalyticsorg.wixsite.com/brain-analytics>

The Brainiac

<https://www.facebook.com/BMEBrainiacs>

Presentations and Posters

Talk – 2025

International

- UK - ASEAN Workshop on Partnership in AI-Powered Health Smart Technologies, Hanoi, Vietnam
- Meeting with J&J leaders, Vietnam

Domestic

- 12th Scientific Conference on diagnostic imaging 2025, Ho Chi Minh City, Vietnam

Talk – 2024

International

- Biomarkers - FORUM: Bridging Neuroscience Discoveries Towards Better Therapeutic Outcomes - Technology in Dementia, 18th International Congress of the Asian Society Against Dementia 2024, Malaysia.
- Intervention Using Cognitive Apps for Patients with Mild Cognitive Impairment (MCI) - Technology in Dementia, 18th International Congress of the Asian Society Against Dementia 2024, Malaysia.
(<https://asad2024.com/programme-detailed>)
- AI Connect II – AI Frontiers: Bolstering Scientific Exchange and Research, Vietnam.
- Webinar IBRO-EEC edition: Latest Advancements in Understanding Alzheimer's Disease - Global Neuroscience Horizons.
(https://www.youtube.com/watch?v=k_JSWWhFdxF4)
(https://drive.google.com/file/d/1bBIzLDrI9WXTFPMFu_5PjLf5k1k83y3A/view?usp=sharing)
- Academic visit to establish a research collaboration that supports comparative studies on living with dementia - University of Hertfordshire.

Domestic

- “Mathematics in Biomedicine”, Math Open Day MOD 2024
- “Reaching for Greater Heights” Talkshow, Auditorium B, Gia Dinh People's Hospital, Vietnam.
(<https://www.facebook.com/share/p/17hpsziHfz/>)
- Scientific research methods course, The Youth Scientific and Technological Promotion Center (TST).
(<https://www.facebook.com/share/p/18yfKDdRJG/>)
- Seminar “Students and scientific research”, Vietnam.
- Seminar “Sharing experiences as a PhD student”, Vietnam.
- Report “Sharing experiences in appointing Professors and Associate Professors in some countries”, Vietnam.

Talk – 2023

International

- Lab meeting - Nanyang Technological University (NTU)
- Neuroengineering approaches for stress detection and intervention in low-middle-income countries, 13th Mind-Body Interface International Symposium
- Rencontres du Vietnam: The first International Symposium of Nano Life Science: Nano Biotechnology, Biosensor, Computation (NanoBioCoM2023), Quy Nhon, Vietnam
- Webinar “Crafting a Successful IBRO Grant Application”
(<https://www.youtube.com/watch?v=-IVldyvVPQI&t=5s>)
(https://drive.google.com/file/d/1-cbNY6orREuDcSSu_kkGBLBQyuBb1K_q/view?usp=sharing)
- Aspects Surrounding Quality of Life, Hyper Interdisciplinary Conference in Vietnam 2023: Advancing Quality of Life: Bridging Human Innovation (<https://global.lne.st/news/ph/2022/12/17/hicvn2023keynote/>)

Domestic

- RMIT's Psychology Career Workshop
- Neuroengineering Approaches for Stress Detection and Intervention in Low-Middle-Income Countries, Signal Processing and Data Science for Smart Health Conference & VISHC Summer School 2023 (VinUniversity)
(<https://smarthealth.vinuni.edu.vn/event/the-first-vishc-summer-school-2023-to-excel-in-research-activities-for-vinuni-undergraduates/>)
- Psychology Research Seminar - RMIT University

Talk – 2022

International

- Women in Neuroscience Symposium
(<https://drive.google.com/file/d/1S8cSfFXgwgkdQuN9GgX3DLwhvEqsd9kV/view?usp=sharing>)
- Application of AI and blockchain in medical and education, International Health Tech Conference.

Domestic

- Multi-Disciplinary Characteristics in Neurology Problems, Math Open Day MOD 2022.
(https://www.facebook.com/watch/live/?ref=watch_permalink&v=1476673259500472)
- Workshop “Gen Z and Scientific research”, International University, Vietnam.
(<https://hcmiu.edu.vn/workshop-chuyen-de-gen-z-nghien-cuu-khoa-hoc>)

Talk – 2021

International

- Effect of alpha wave-containing music on the psychological stress level of hospital staffs and their cognitive performance - The 3rd IEEE Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (Best Paper Award).
- Investigating brain and cardiovascular responses to develop a machine learning model for acute and chronic stress detection - International Biomedical Science and Engineering Symposium, Gwangju Institute of Science and Technology, November 25-26, 2021.
- World Dementia Council
(<https://drive.google.com/file/d/1fd5GjcC9olmurbcMrrPptxZlRWwnRh-h/view?usp=sharing>)

Domestic

- Workshop “Finding creative ideas about scientific research”, The Youth Scientific and Technological Promotion Center (TST).
(<https://khoahoctre.com.vn/workshop-tim-kiem-y-tuong-sang-tao-ve-nghien-cuu-khoa-hoc-2/>)
- The Future is Female - Fulbright University Vietnam Career Day 2021
- Guest lecture: Research methodology and application of neuroscience - Vietnam Biology Olympiads team, High School for the Gifted – VNU-HCM
- Characteristics and steps to conduct scientific research - Ho Chi Minh Communist Youth Union, Vietnam
- Seminar “Women in Innovation and Creativity”.
(https://www.dropbox.com/scl/fi/lm87grzx6hbz38fiqqeah/Huong_Women-in-Innovation-Panelist.jpeg?rlkey=1dhmdoojoevbom6ed8q5groj6&st=sgiw167t&dl=0)

Talk – 2020

International

- SEED: A Framework for Stress Estimation Using Emotiv Devices, The 19th International Conference on New Trends in Intelligent Software Methodologies, Tools and Techniques, SoMeT 2020, Kitakyushu, Japan, September 22-24, 2020.
- Brain activity associated with examination-induced stress in college students, The 2nd University of Tsukuba-University of Science Joint Young Researcher Meeting in Biomedical Sciences, The 1st Student Conference, University of Science.

Domestic

- Application of Artificial Intelligence to develop tools for diagnosing Alzheimer’s Disease, The 2nd Symposium on Alzheimer’s Disease at University of Medicine and Pharmacy, Ho Chi Minh city, Vietnam.
- Towards a non-invasive diagnosis for Alzheimer’s Disease, The 1st Symposium of the Vietnam Alzheimer Network, National Geriatric Institute, Ho Chi Minh city, Vietnam.
- Lecture: A research methodology course, I-IVY education, Ho Chi Minh, Vietnam

Talk – 2019

International

- Perspective: Brain Aging in Vietnam: Society needs and research approach, The 9th Neuroscience International Symposium: Bridging Brains – Enriching Minds 2019 in the Philippines

Domestic

- Introduction into Mindfulness Based Stress Reduction, International University, Vietnam National University, Ho Chi Minh City, Vietnam.
- Women in Science, Unfold the V, Ho Chi Minh City, Vietnam.
- The common myths and the neurobiology of autism, Hoa Sen University, Ho Chi Minh City, Vietnam.
- Developing diagnostic methods for neurodegenerative diseases, Symposium on Laboratory Test and Quality Control, University of Medicine and Pharmacy, Ho Chi Minh city, Vietnam.
- Workshop “Working with scientific posters” (<https://www.facebook.com/share/p/1DS06E7magg/>)

Talk – 2018

- EEG applications in Alzheimer’s Disease Detection, Alzheimer’s Disease Awareness Day Symposium, University of Medicine and Pharmacy Medical Center, Ho Chi Minh City, Vietnam

Poster – 2023

- Comparing Hippocampus Volume in Alzheimer’s Disease and Healthy Controls: A MRI Analysis with HarP T
- Analyzing the effect of stimulation frequency on the performance of Machine Learning Classification on public Migraine EEG dataset
- Plasma cell-free RNA Profiling of Vietnamese Alzheimer’s patients: A pilot study
- The Role of Quantitative EEG in Assessment of Cognitive Changes after Ischemic Stroke
- Evaluating stress responses in Vietnamese university students to Trier Social Stress Tasks using multiple measures as a reference for a Point-of-care stress diagnostic device

Poster – 2022

- Development and Commercialization of a Brain Training App Targeting the Vietnamese Elderly

Poster – 2021

- Current state of Alzheimer's diagnosis in Vietnam and doctors' view of an Alzheimer's AI detector
- Immediate effects of Mindfulness-Based Stress Reduction
- Early diagnosis of Alzheimer's disease using Extreme Gradient Boosting (XG-Boost) on MRI images
- Development of a Cortisol-quantification test strip for stress evaluation using low-cost materials
- Mindfulness-based Stress Reduction program for college students: Effect on stress level and EEG power
- Investigating brain and cardiovascular responses to develop a machine learning model for acute and chronic stress detection

Poster – 2018

- Shank and zinc mediate an AMPA receptor subunit switch in developing neurons
The Cold Spring Harbor Conference in Osaka, Japan
The Molecular and Cellular Neurobiology Gordon Conference in Hong Kong.
The Bay Area Molecular and Cellular Neuroscience meeting, California, USA.

C. Teaching experience

Current courses:

Undergraduate program

1. Brain Computer Interface (BCI) - 1 class/term & 1 term/year - 4 credits (Technical Elective course)
2. Principles of Neuro-engineering - 1 class/term & 1 term/year - 4 credits (Technical Elective course)
3. Bioethics - 1 class/term & 2 terms/year - 3 credits (Foundation course)
4. Project 1B: in vitro studies - 2 classes/term & 2 terms/year - 1 credit (Foundation course)

Graduate program

1. Research Methodology - 1 class/term & 2 terms/year - 3 credits
2. Progress in Biomedical Engineering - 1 class/term & 2 terms/year - 3 credits
3. Engineering Challenges in Medicine - 1 class/term & 2 terms/year - 3 credits
4. Advanced Biosignal Processing - 1 class/term & 2 terms/year - 2 credits
5. Computational Neuroscience and Applications - 1 class/term & 1 term/year - 2 credits
6. Biostatistics - 1 class/term & 1 term/year - 3 credits

Previous teaching experience:

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|---|-----------|
| (1) Coach at Research and Design workshop, Stanford D-School | 2014 |
| (2) Lead Instructor at Stanford School of Medicine Health Matters, Brain Day and RISE programs | 2013-2014 |
| (3) Teaching Assistant at Stanford Intensive Neuroscience Course | 2012 |
| (4) Teaching Assistant in the General Molecular Biology Techniques at University of Sciences- Vietnam | 2011 |

D. Positions and Awards

Positions and Employment

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| Educational Quality Assurance Coordinator, School of Biomedical Engineering, International University, HCMC VNU | 2021 – present |
| Chair, Department of Tissue Engineering and Biomedical Engineering, School of Biomedical Engineering, International University, HCMC VNU | 2020 – present |
| Research coordinator, School of Biomedical Engineering, International University, HCMC VNU | 2019 – 2020 |
| Lecturer, School of Biomedical Engineering, International University, HCMC VNU | 2018 – present |
| Psychiatry Department, School of Medicine, Stanford University | 2014 – 2015 |
| Department of Neurology and Neurological Sciences, School of Medicine, Stanford University | 2012 – 2014 |
| Research assistant at the Oxford University Clinical Research Unit (OUCRU) | 2011 – 2012 |

Other Experience and Professional Memberships

The Rising Talents List of the Precision Nanomedicine Editorial Board	2024 – present
Member of IBRO Early Career Committee	2024 – present
Member of the National Academies' Workshop Planning Committee on	2023 – present
Engaging Scientists in Shared Responsible Innovation in Neuroscience in Southeast Asia	
Member of editorial board, Brain Behavior and Immunity Integrative	2023 – present
Guest Editor, IBRO Neuroscience Report Mini-series for the IBRO APRC Alumni	2021 – present
Member, Brain connects (international organization)	2019 – present
Organizing Committee Member, BrainConnect (which includes supporting the Special Research Topic on Frontier Human Neuroscience)	2019 – present
Vice chair, Bioethics committee, School of BME, International University, HCMC VNU	2019 – 2021
Advisory board, International School at Park City Hanoi (ISPH), Vietnam	2018 – present
Member, Biomedical Engineering (BME) society, Vietnam	2018 – present
Student Program Committee Member, Stanford Neurosciences Graduate Program	2015 – 2018
Reviewer (Journal of Neuroscience, Nature Communications, Neuroscience, Neuron, Vietnam Journal of Science, PNAS)	2015 – present
Leader, Vietnam Book Drive project, VEFFA	2013 – 2014
Executive Member, Vietnam Education Foundation Fellow Association (VEFFA)	2013 - 2014
Managing Editor and Founder, Vietnam Journal of Science	2013 – 2017

Individual Awards

Second Prize of the 5th Vietnam Medical Achievement Award, Vietnam.	2025
Vietnamese Women	2024
Advanced Youth of Ho Chi Minh City following Uncle Ho's teachings	2024
Outstanding achievements in studying and following Ho Chi Minh's ideology, morality and style in the period 2021-2024	2024
Outstanding female intellectual of Ho Chi Minh City 2019-2024, Vietnam	2024
Golden Globe Science and Technology Award, Vietnam	2023
National Outstanding Teacher	2023
Outstanding Young Citizen of Ho Chi Minh City	2023
Women of the Future Awards Southeast Asia	2023
Vietnamese Outstanding Young Face Award	2023
Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada	2023
Second Prize, China-ASEAN Innovation and Entrepreneurship Competition	2023
International Awards L'Oréal-UNESCO for Women in Science	2022
Top 4 Social Business Creation, HEC Canada	2022
Third Prize, National Technological Creation Contest, Vietnam	2022
Second Prize, Ho Chi Minh City Creative Awards	2021
SUN* Award, Tech Planter Demo day in Vietnam	2021
Top 5 Business Innovation Idea, Canada	2021
Staff Excellent Award, International University, VNU, Ho Chi Minh City, Vietnam	2021
Young Talent Award, Vietnam National Youth Association	2020
Top 5, BAMBUuP Disruption in Health Care, Vietnam	2020
Top 8, Ho Chi Minh City Artificial Intelligence Competition, Vietnam	2020
Early Career Award, International Brain Research Organization	2020
ALBA-FKNE-YIBRO Diversity Grant for participating in the FENS 2020 Virtual Forum	2020
Travel Grant to Attend Science Conference (Stanford University Office of Graduate Education)	2018
Faculty for the Future Fellowship (The Schlumberger Foundation, Paris, France)	2015 - 2018
Outstanding Service Award (The Vietnam Education Foundation Fellow Association - VEFFA, US)	2014
Neurobiology Course Summer Scholarship (The Marine Biological Laboratory, Massachusetts, US)	2014
Vietnam Education Foundation Fellowship (The Vietnam Education Foundation, Virginia, US)	2012 – 2014
Stanford Graduate Fellowship (Stanford University, California, US)	2012 – 2015
Five Goods Award (The People Committee, Ho Chi Minh City, Vietnam)	2012
Eureka Scientific Award (The Center of Technology and Development, Ho Chi Minh City, Vietnam)	2011
Award for Exceptional Graduation (The University of Science, Ho Chi Minh City, Vietnam)	2011
Scholarships for outstanding high school and college students (The Odon Vallet Foundation, Sumitomo Corporation, and Lawrence S. Ting Memorial Funds).	2006 – 2011

Awards from my lab students

(2024) AmCham Scholarship Program.

(2024) Pony Chung Scholarship.
 (2024) Vingroup Scholarship – VinUniversity.
 (2024) Travel award NEURO2024 Conference.
 (2024) IBRO-APRC Associate School on “Neuro-Enhancement for Brain Health”.
 (2024) First Prize, Scientific Conference “Application of Digital Transformation and Artificial Intelligence in Disease Diagnosis and Treatment.”
 (2024) Silver Medal in the “Design, manufacture, application” award, Vietnam.
 (2023) IUPAB Travel Award, Rencontres du Vietnam: The First International Symposium of Nano Life Science: Nano Biotechnology, Biosensor, Computation (NanoBioCom2023), Vietnam.
 (2023) 5-min Poster Blitz, 13th Mind-Body Interface International Symposium.
 (2023) Travel award, 13th Mind-Body Interface Symposium, Taiwan.
 (2023) Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada.
 (2023) Second Prize, China-ASEAN Innovation and Entrepreneurship Competition.
 (2023) Round 4 and the Grand Final Week, Social Business Creation, Montreal, Canada.
 (2023) Third prize in the contest "Digital Transformation - Creativity Challenge", Ho Chi Minh City, Vietnam.
 (2023) ADAI Fellowship, Ho Chi Minh City, Vietnam.
 (2022) UNLEASH HACKS: Mental Health Europe.
 (2022) Top 4 Social Business Creation, HEC Canada.
 (2022) Third Prize, National Technological Creation Contest, Vietnam.
 (2022) ADAI Fellowship, Ho Chi Minh City, Vietnam.
 (2022) ITR Graduate Fellowship, Ho Chi Minh City, Vietnam.
 (2021) Second Prize, Ho Chi Minh City Creative Awards.
 (2021) SUN* Award, Tech Planter Demo day in Vietnam.
 (2021) Top 5 Business Innovation Idea, Canada.
 (2020) Top 5, BAMBUuP Disruption in Health Care, Vietnam.
 (2020) Top 8, Ho Chi Minh City Artificial Intelligence Competition, Vietnam.

E. Funding

<i>No.</i>	<i>Title of project</i>	<i>ID number & Level of management</i>	<i>Duration</i>	<i>Cost (EUR)</i>	<i>Role</i>	<i>Achieved Outcome</i>
1	An ultra-sensitive plasma p-tau 217 immunosensor using screen-printed carbon electrodes modified with MWCNTs/metallic nanoparticles and oriented immobilized antibody for Alzheimer’s Disease detection		2024-2025		Principal investigator	
2	Multimodal Screening & Management System of Mental Wellbeing in Low Resource Settings		2023-2024		Co-principal investigator	
3	BrainConnects 2023 Conference Grant		2023	6000	Host	
4	Investigation and Implementation of Solutions for Challenges in	Medical Research Council,	2020-2021	25000/ 1 year	Main investigator	1 international conference

	Mental Issues of the Elderly in Vietnam by Networking with Advanced Countries	United Kingdom				
5	Toward a non-invasive blood test for Alzheimer Disease	Vietnam Alzheimer Network	2020–2022	10000/ 2 years	Principal investigator	01 international conference publication
6	Grant for Outstanding Research Group	#NCM2020-28-01 HCMC VNU	2020 - 2025	3800/ year	Main investigator	01 international conference publication 01 international conference talk
7	Development and Evaluation of a Cortisol quantification test strip for reliable and low-cost stress diagnosis.	C2022-28-01	2021 - 2023	3000	Principal investigator	01 Q1-paper publication 01 international conference publication
8	Investigation on establishing a large database of EEG and video recordings of Vietnamese people in application for intelligent control and primary motor rehabilitation in epileptic patients	# KC-4.0-07/19-25, Ministry of Science and Technology	2020-2023	19000	Main investigator	01 international conference publication
9	Investigating a model of Tele-EEG coupled with ultrasound function emitter on stress evaluation of college students	# KC-4.0-07/19-25, HCMC VNU	2020-2022	3800	Main investigator	01 international conference publication 01 international conference talk
10	Artificial Intelligent system for automatic diagnosis of Alzheimer's disease using neuroimaging	#T2019-05-BME, Internal grant of International University, HCMC VNU	2019 - 2021	1900	Main investigator	01 international conference publication
11	Investigate the effects of mindfulness-based stress reduction (MBSR) program on stress level and brain activity of college students	#T2019-04-BME Internal grant of International University, HCMC VNU	2019-2021	1900	Principal investigator	01 international conference publication 01 international conference talk
12	Solid-state patch clamp platform to diagnose	#5R33MH100717-04, National	2013-2018		Student Participant	01 Q2-paper publication

	autism and screen for effective drug	Institute of Health, US				
13	Astrocytic Control of GABA Inhibition in Epilepsy	#5R01NS090911-04, National Institute of Health, US	2016-2018		Student Participant	01 Q1-paper publication

E. REFERENCE

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